



# Cloud Solution

for

# Polytronic Training Systems

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## Core Objectives - Cloud Principles and Design

### Business Research

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|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Business Name                             | <b>Polytronic Training Systems LLC</b>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Number of Employees                       | 100 Employees                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Location(s)                               | Tawazun Industrial Park, Al Ajban, Abu Dhabi, United Arab Emirates                                                                                                                                                                                                                                                                                                                                                                                                           |
| How long has the company been in business | <b>Polytronic Training Systems LLC</b> has been in business for more than 7 years in UAE since 2017.                                                                                                                                                                                                                                                                                                                                                                         |
| Purpose of the Business                   | <b>Polytronic Training Systems LLC</b> provides electronic targetry equipment, training services, and installs electronic targetry systems and ranges for both military and non-military applications.                                                                                                                                                                                                                                                                       |
| Existing Technologies                     | The organization currently utilizes an on-premises data center to host sensitive data, including client's records and financial systems. For cloud-based services, they have adopted Microsoft 365 for internal email communication and shared department resources.                                                                                                                                                                                                         |
| Other Notes                               | <b>Polytronic Training Systems LLC</b> , located in Abu Dhabi, UAE, has been in operation since 2017 with 100 employees. The company provides electronic targetry equipment and training services for both military and non-military -applications. It uses an on-premises data center for sensitive data and Microsoft 365 for internal communications. Future plans may include enhancing IT infrastructure to support growing operational needs and ensure data security. |

## Deployment Model

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For **Polytronic Training Systems LLC**, the **Hybrid Cloud** deployment model is the most suitable choice. This decision is based on the company's current setup of using an on-premises data center for sensitive data and Microsoft 365 for email communications, indicating the existing use of both cloud and local infrastructures.

A hybrid cloud model allows the business to maintain control over its sensitive data by keeping it on-premises while taking advantage of the scalability and flexibility offered by cloud services for non-sensitive operations. This approach supports future scalability needs, enables efficient resource utilization, and enhances data security while meeting the company's requirements for flexibility in deploying new services.

## Service Model

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For the cloud service model, **Infrastructure as a Service (IaaS)** is recommended for **Polytronic Training Systems LLC**. The organization already relies on an on-premises data center, indicating a preference for managing its infrastructure. IaaS offers a cost-effective way to extend or migrate their existing infrastructure to the cloud without giving up control over servers, storage, and networking components.

By using IaaS, **Polytronic Training Systems LLC** can deploy virtual machines, storage, and networking resources on demand, reducing the need for costly hardware investments. Additionally, it provides the flexibility to scale resources as needed while retaining control over the configuration and management of the environment, supporting their plans to enhance IT infrastructure and improve data security.

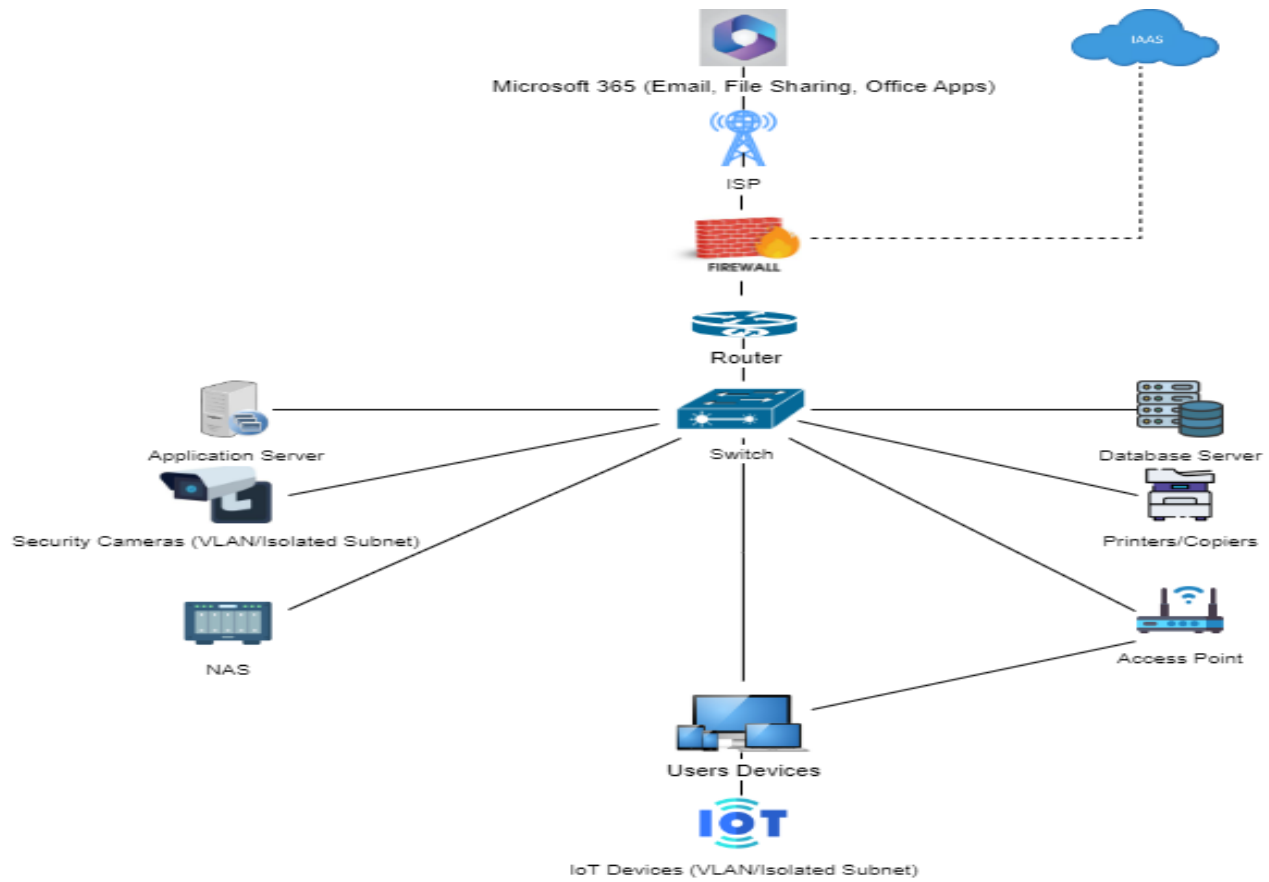
## Cloud Design for your Business

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The proposed cloud design for **Polytronic Training Systems LLC** focuses on ensuring redundancy, high availability, and disaster recovery. To achieve redundancy, multiple virtual machines and storage resources will be deployed across different data centers in the cloud. This setup will prevent single points of failure, ensuring that if one component fails, others can take over to maintain system operations.

High availability will be implemented using load balancers to distribute traffic evenly between the on-premises servers and cloud instances, minimizing downtime and ensuring smooth service delivery. For disaster recovery, regular backups of critical data will be conducted, and a failover system will be established. This plan will allow quick recovery of essential services in the event of hardware failures or cyber incidents, helping maintain business continuity.

## Infrastructure Diagram



# Implementation Phases - Cloud Networking and Storage

## Additional Business Research

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|                                |                                                                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Internet Service Provider Name | Etisalat Business Pro 550Mbps                                                                                                                                                 |
| Data Connection Type           | Cat 6e connection type provides high-speed internet with low latency, suitable for both on premises and cloud-based operations.                                               |
| Download and Upload speeds     | Download: 250 Mbps, upload 200 Mbps, for maximum of 50 users, as other employees work remotely or in the fields.                                                              |
| More than one ISP?             | No, <b>Polytronic Training Systems LLC</b> uses Etisalat as the sole ISP for internet connectivity ensuring reliable service for both on-premises and cloud-based operations. |

## Procedures when ISP goes down.

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Notify IT support team to acknowledge the outage and start troubleshooting.  
Confirm if the issue is specific to the building or network-wide by testing multiple devices.  
Contact Etisalat's customer support to report the outage and receive updates on resolution times.  
If the outage lasts longer than an hour, switch to a backup internet solution (e.g. mobile hotspot).  
Document the event, including time of downtime, affected services, and steps taken for future reference and analysis.  
Inform employees about the situation and provide alternatives for communication (e.g. mobile devices, email forwarding).

## Virtual Private Network

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A Virtual Private Network (VPN) is used to securely connect remote employees to the business network, especially for sensitive operations like handling client data and financial records.  
The VPN uses encryption protocols (e.g. SSL or IPsec) to protect data in transit.  
Remote workers are required to authenticate with multi-factor authentication (MFA) to strengthen security.  
VPN access is granted based on employee roles and the necessity of accessing internal systems remotely.  
Usage of the VPN is monitored and logged to ensure compliance with security policies and industry regulations.

## Kinds of Files Stored

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Client Records: Sensitive data related to clients, including contracts, training schedules, and feedback.

Financial Systems: Accounting files, payroll records, and tax related documents.

Project Files: Technical specifications, reports, and blueprints for custom targetry systems.

Internal Documents: Company policies, HR records, and internal communications stored in Microsoft 365.

Marketing Materials: Brochures, training manuals, and product catalogs.

Backup Data: Regular backup copies of essential operational data, both on-premises and in the cloud, for disaster recovery.

## How will the Business Connect?

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**Polytronic Training Systems LLC** will use secure VPN connections for remote employees to access the company network and cloud-based services, ensuring safe and reliable access to sensitive data and systems.

Internal communications will occur via Microsoft 365, providing email, shared documents, and collaboration tools.

Video conferencing solutions like Microsoft Teams or Zoom will be used for remote meetings and training sessions with clients.

Cloud storage services (such as OneDrive and Azure) will be used for file sharing and collaboration on project files, ensuring seamless integration between on-premises and cloud systems.

On-premises connectivity (LAN) will be used for internal operations within the office, allowing quick access to local systems and databases.

## Domain Name System

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**Polytronic Training Systems LLC** uses third-party DNS provider for managing their domain names and ensuring high availability of services.

The DNS is configured with redundancy in case of service interruptions to ensure consistent access to the company's web services.

Security measures such as DNSSEC (DNS Security Extensions) are implemented to prevent DNS hijacking or poisoning.

Regular DNS audits are conducted to ensure accuracy and prevent misconfigurations.

Failover DNS services are set up to minimize downtime if the primary DNS provider experiences issues.

## Storage Providers

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**Polytronic Training Systems LLC** utilizes a hybrid cloud storage strategy, combination both on-premises storage (for sensitive data) and cloud storage (for less sensitive data and backups).

Microsoft OneDrive and Azure are used for cloud storage and file sharing across departments.



On-premises storage is used for backup copies of critical systems and sensitive data, including client records and financial systems.

Local NAS or SAN storage is implemented for high-speed access for frequently used files and to provide redundancy.

Data is encrypted both at rest and in transit to ensure security compliance and protect against data breaches.

## Content Delivery Network

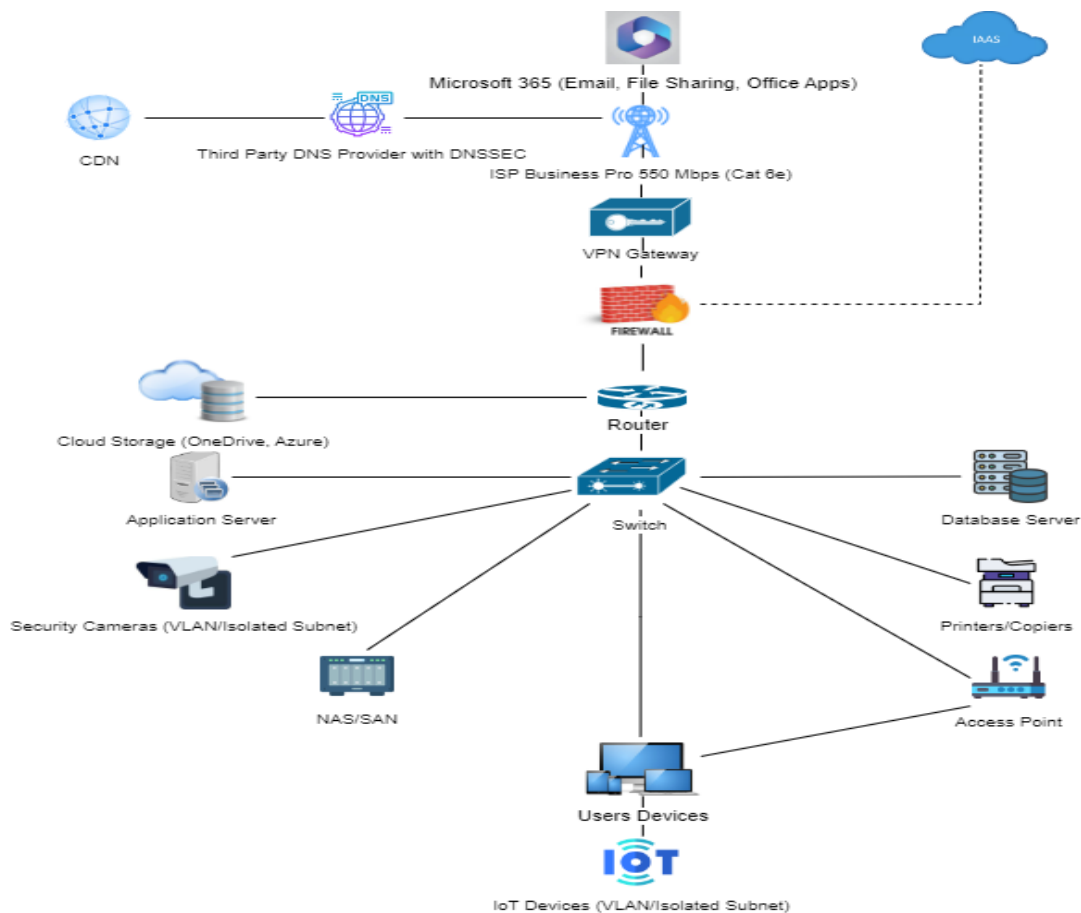
Polytronic Training Systems LLC utilizes CDN (Content Delivery Network) for optimizing web content delivery to users.

The CDN caches static resources such as training videos, product documentation, and marketing materials across global edge servers to reduce latency and improve access to speed. The CDN also supports high availability and scalability during peak usage times, ensuring minimal disruption to online training sessions or web-based services.

Security features, such as DDoS protection, are implemented through the CDN to protect the company's public-facing resources from malicious attacks.

The CDN integrates with the company's website and other services to improve performance for customers globally, ensuring smooth user experience.

## Updated Infrastructure Diagram



# Needs Assessment – Assessing Cloud Needs

## Research

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Key Stakeholders                            | <p><b>CEO:</b> Aligns cloud adoption with the company's long-term vision and goals.</p> <p><b>CFO:</b> Evaluates the financial impact, focusing on Return on Investment (ROI) and cost efficiency.</p> <p><b>CTO:</b> Oversees the technical integration of cloud services and infrastructure.</p> <p><b>IT Manager:</b> Manages daily operations, monitors cloud performance, and handles incident response.</p> <p><b>Department Heads:</b> Provide insights on specific needs and pain points for cloud service adoption</p> |
| Single Point of Contact for each department | <p><b>Sales Department:</b> Sales Manager for cloud-based Customer Relationship Management (CRM) and sales applications.</p> <p><b>Finance Department:</b> Finance Manager for cloud Enterprise Resource Planning (ERP) and accounting tools.</p> <p><b>IT Department:</b> IT Manager for cloud service management and support.</p> <p><b>Operations Department:</b> Operations Manager for cloud-based logistics and inventory solutions.</p>                                                                                  |
| SPOC for the organization                   | <p><b>Chief Technology Officer:</b> The primary point of contact for all cloud-related initiatives and escalations.</p>                                                                                                                                                                                                                                                                                                                                                                                                         |

## Business

### Benefits of Cloud Adoption:

- Improved efficiency and reduced operational costs.
- Enhanced business continuity and disaster recovery capabilities.
- Scalability to handle increased demand.
- Better customer experience through faster service delivery.

### Needs for the Next 6-12 Months:

- Adopting a cloud-based Customer Relationship Management (CRM) system.
- Increased storage capacity to manage growing data volumes.
- Enhance remote work capabilities for workforce flexibility.

## Technical

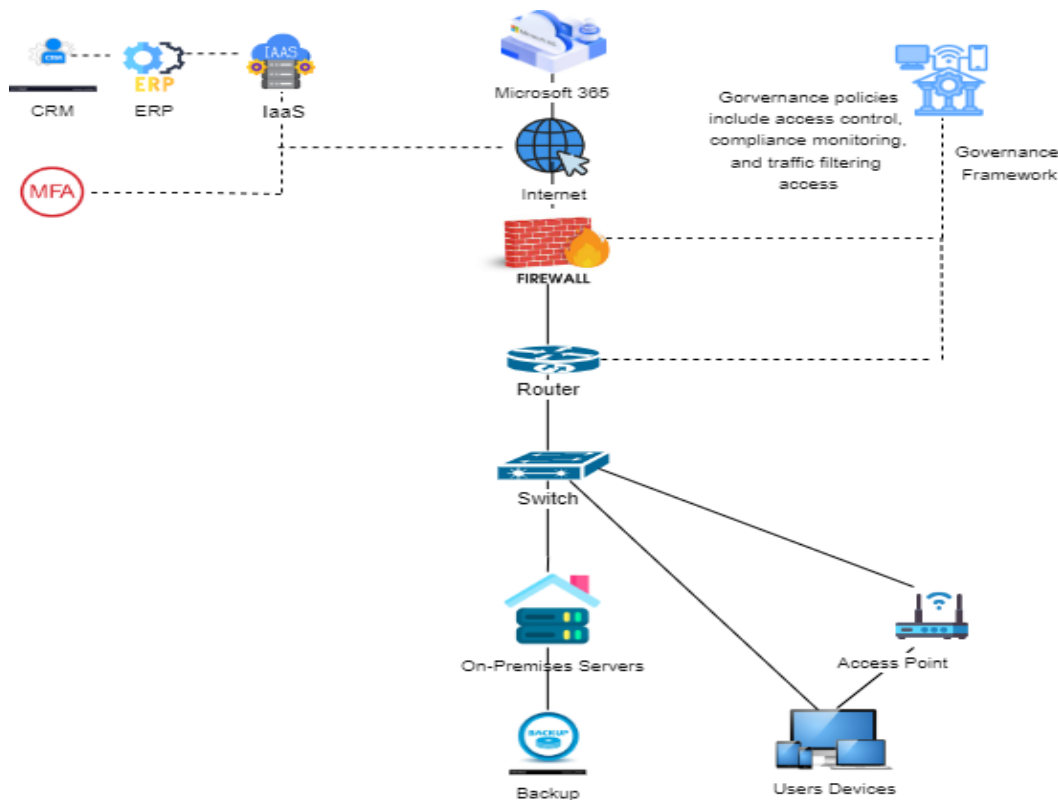
### Current Needs:

- Migration of on-premises applications to a cloud platform for better accessibility.
- Implementation of robust data security measures, including encryption and access controls.
- Integration of exiting on-premises IT systems with cloud infrastructure to ensure seamless operations.

### Future Needs:

- Scalable solutions to support business expansion and new workloads.
- Advanced monitoring tools for proactive systems performance management.
- Machine learning and AI capabilities to improve data insights and automation.

## Framework Diagram



## Baseline

### Current Performance:

Existing on-premises infrastructure performs adequately but encounters latency during peak usage, especially with large file transfer or high concurrent access.

**Cloud Usage:**

Utilizes Microsoft 365 for email and document sharing but lacks advanced cloud tools for CRM and ERP, and backups.

**Feasibility**

**Cloud Requirements:**

Adoption of CRM/ERP systems to streamline customer management and operational efficiency.  
Cloud backup solutions to supplement local NAS and improve disaster recovery.

**Integration Capabilities:**

Hybrid cloud integration using existing firewall and router configurations for secure and encrypted access to cloud resources.

**Gap Analysis**

**Polytronic Training Systems LLC** currently faces challenges with high costs, inefficient business operations, and limited functionality due to its reliance on on-premises infrastructure. Key gaps include inadequate hybrid cloud integration, inconsistent security protocols, limited employee expertise with cloud technologies, and a lack of formal governance frameworks for cloud adoption.

By addressing these gaps through targeted improvements in technology, training, governance, and security, the organization can successfully transition to a scalable and secure IaaS hybrid cloud environment. This approach will enhance operational efficiency, reduce costs, and empower employees to utilize cloud services effectively.

**Gap Analysis Diagram**

| Category | Current State                    | Goal                                        | Action Needed                            | Priority | Owner      | Due Date     |
|----------|----------------------------------|---------------------------------------------|------------------------------------------|----------|------------|--------------|
| Business | High costs and limited cloud use | Increase cloud adoption for storage and CRM | Implement cloud CRM and backup solutions | High     | CTO        | Next Quarter |
| People   | Limited cloud tool knowledge     | Improve skills through training             | Conduct training sessions                | Medium   | HR Manager | 2 Months     |

|                   |                                  |                                     |                                                    |        |                    |             |
|-------------------|----------------------------------|-------------------------------------|----------------------------------------------------|--------|--------------------|-------------|
| <b>Governance</b> | Basic cloud policies             | Develop a formal framework          | Create and enforce governance polices              | Medium | IT Manager         | 1 Month     |
| <b>Platform</b>   | Minimal hybrid cloud integration | Enable full integration             | Implement hybrid environment setup                 | High   | IT Manager         | 3 Months    |
| <b>Security</b>   | Inconsistent protocols           | Standardize security across systems | Establish unified protocols with encrypted and MFA | High   | Security Officer   | Immediate   |
| <b>Operations</b> | Limited cloud operational tools  | Streamline logistic and inventory   | Deploy ERP-integrated cloud tools                  | High   | Operations Manager | Next Quater |

# Vendor Selection Process – Engaging Cloud Vendors

## Capital and Operating Expenditures

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### Capital Expenditures (CapEx)

**On-Premises Server Upgrade:** This is a one-time investment in upgrading existing servers to improve performance for hybrid cloud integration. This cost is necessary to ensure compatibility with cloud resources and reduce latency.

**Networking Equipment:** This involves the acquisition of advanced firewalls and switches to facilitate secure hybrid cloud operations and encrypted data transmission.

### Operation Expenditures (OpEx)

**Cloud Subscription Fees:** These are ongoing costs for using CRM/ERP software as a service (SaaS), which scales with usage and eliminates the need for upfront investment.

**Technical Support and Maintenance:** Regular payments for cloud vendor support plans to ensure smooth operations and prompt issue resolution.

## Potential Cloud Vendors

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### Amazon Web Services (AWS)

Offers a broad range of cloud solutions, including hybrid integration, CRM, and ERP support. Provides robust security features, scalability, and global data center coverage.

### Microsoft Azure

Seamless integration with existing Microsoft 365 tools and on-premises systems. Strong focus hybrid cloud environment and advanced analytics tools.

### Google Cloud Platform (GCP)

Known for its data analytics and machine learning capabilities, making it a good option for companies focused on advanced analytics.

Provides a secure and highly scalable cloud environment with deep integration with Google's services and open-source technologies.

## Variable and Fixed Costs

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### Variable Costs

**Cloud Storage Usage:** Costs increase as storage needs grow with company expansion or seasonal demand.

**Data Transfer Costs:** Expenses fluctuate based on the volume of data transferred between the cloud and on-premises systems.

### Fixed Costs

**CRM subscription Fee:** A consistent monthly fee for a fixed number of users, regardless of usage.

**Cloud Support Plan:** Fixed yearly subscription for 24/7 technical support, independent of issue frequency.

## Licensing Model

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### Subscription-Based Licensing

For the CRM and ERP systems, a pay-as-you-go model provides flexibility and scalability. Licenses are priced per user, with options to scale up or down the workforce changes.

### Bring Your Own License (BYOL)

BYOL allows companies to transfer their existing on-premises software licenses to cloud environments. This model helps businesses save costs by avoiding the need to purchase new licenses for cloud usage.

## Evaluation of Cloud Vendors

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### Evaluation Type: Pilot Implementation

A pilot was conducted with AWS with Azure for subnet operations (CRM and hybrid cloud integration).

**AWS:** Demonstrated high reliability and efficient hybrid connectivity, but slightly higher costs compared to Azure.

**Azure:** showcased seamless integration with existing Microsoft tools, making it easier for staff to adapt.

**GCP:** Excelled in advanced analytics and machine learning tools but required more expertise for optimal use.

### Performance Metrics

**AWS:** Faster response times to data retrieval by required additional customization for integration.

**Azure:** Slower performance during high-concurrency operations but excelled in data analytics capabilities.

**GCP:** Showed excellent performance in handling large-scale data analytics workloads with minimal latency.

### Choosing the Right Cloud Solution

**Polytronic Training Systems LLC** must assess whether a public, private or hybrid cloud model best fits its business needs. Companies with large IT infrastructures may opt for a private cloud or combine both open-source and proprietary solutions for different purposes.

### Gathering Information

The initial step for **Polytronic Training Systems LLC** is to research and request information from potential vendors to compare their offerings. This process is crucial for identifying the right fit to achieve the expected savings and quality.

## Open Source vs. Proprietary Technology

**Polytronic Training Systems LLC** Must decide between open-source and proprietary solutions. Open-source solutions generally offer lower costs and greater flexibility, while proprietary solutions provide stability and extensive support.

## Request for Information (RFI)

**Polytronic Training Systems LLC** should issue an RFI to collect detailed written information from cloud service providers regarding their capabilities. This document will help guide the vendor selection process by providing essential insights into each provider's offerings.

## Request for Quotation (RFO)

**Polytronic Training Systems LLC** is encouraged to issue an RFQ to gather precise pricing details for a set of services after completing the RFI. This will help identify cost-effective solutions while ensuring vendors understand the organization's technical requirements.

## Statement of Works (SOW)

A SOW should be included as a foundational document for defining the scope, deliverables, and responsibilities of both **Polytronic Training Systems LLC** and the selected vendor. The SOW serves as critical part of the vendor relationship, ensuring accountability and clear expectations throughout the project life cycle.

## Request for Proposal (RFP)

To address broader organizational goals and complex operational needs, **Polytronic Training Systems LLC** should initiate an RFP process. This formal process requests vendors to propose comprehensive solutions tailored to the company's requirements.

## Service Level Agreement

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### AWS SLA

**Performance Levels:** 99.95 uptime guaranteed.

**Recompense:** Offers service credits for downtime exceeding SLA commitment.

**Outage Handling:** Detailed incident response and compensation structure.

### Azure SLA

**Performance Levels:** 99.95% uptime guaranteed.

**Recompense:** Service credits issued for outage beyond SLA terms.

**Outrage Handling:** Proactive notifications and customer support during downtime events.

### GCP SLA

**Performance Levels:** 99.95% uptime guaranteed.

**Recompense:** Offers service credits for significant outages beyond SLA commitments.

**Outage Handling:** Provides a clear support escalation process downtime and detailed incident tracking.



## Egress Terms

Addressing egress terms is crucial to ensuring **Polytronic Training Systems LLC** retains control over the data if the contract with the cloud provider (CSP) ends.

# Operational Framework – Management and Technical Operations

## Migration Principles

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### Migration Strategies

**Polytronic Training Systems LLC** should choose the most suitable migration strategy based on their needs:

**Incremental Migration:** Start by migrating non-critical systems to the cloud to reduce risk and minimize disruption to core business operations.

**Hybrid Cloud Approach:** Retain sensitive data on-premises for enhanced security while for scalability, flexibility, and efficient backup solutions.

**Employee Training:** Organize interactive workshops and training sessions to help staff adapt to the new tools, processes, and workflows effectively.

### Lift and Shift

Moving applications and data to the cloud without modification, which is quick and easy, but may not optimize cloud benefits.

### Rip and Replace

Completely redesign applications using cloud-native components for optimized performance and cost savings.

### Hybrid Migration

Combine lift-and-shift and rip-and-replace approaches or keep some infrastructure on-premises while migrating other components to the cloud.

### Phased Migration

Migrate data and application in stages, which helps reduce risks and allows for testing and adjustments throughout process.

### Managing Vendor Lock-in

Vendor lock-in is a significant risk when adopting cloud technologies, as it can make transitioning between providers costly and complex. This occurs due to reliance on proprietary technologies, data formats, or restrictive long-term contracts. **Polytronic Training Systems LLC** must consider this challenge as part of its cloud migration and vendors selection strategy.

## Aspects of Operating

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### Data Management: Replication, Locality, and Backup

Establish a multi-region replication strategy to ensure data availability and improve data access for customers based on their locations. Schedule automated daily backups through the cloud service provider's (CSP)

|                                                        |                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                        | managed backup services to minimize data loss and enable swift disaster recovery.                                                                                                                                                                                                         |
| <b>Availability: Zone &amp; Geo-Redundancy</b>         | Employ zone-redundant storage and geo-redundant disaster recovery solutions to maintain high service availability. These measures ensure operational continuity during natural disasters or data center failures.                                                                         |
| <b>Disposable Resources</b>                            | Leverage ephemeral resources for short-term workloads. Virtual machines (VMs) and similar resources will be automatically provisioned as needed and decommissioned once tasks are completed, reducing costs and enhancing resource efficiency.                                            |
| <b>Monitoring and Visibility: Alerts &amp; Logging</b> | Implement robust monitoring tools such as AWS CloudWatch or Azure Monitor to track resource utilization and system health. Configure real-time alerts to notify IT teams of critical incidents. Logs will be securely archived to support compliance, auditing, and debugging activities. |
| <b>Optimization: Auto-scaling &amp; Right-Sizing</b>   | Deploy auto-scaling capabilities to dynamically allocate resources based on workload demands. Conduct periodic audits to ensure VMs, databases, and other resources are correctly sized, eliminating waste and optimizing operational costs.                                              |

## Development & Operations

### Provisioning: Infrastructure as Code(IaC) and Templates

Adopting Infrastructure (IaC) tools such as Terraform or AWS CloudFormation allows organizations like **Polytronic Training Systems LLC** to define and automate infrastructure configurations within cloud environments. This approach ensures consistent deployments across development, testing, and production environments while minimizing manual intervention and errors. For Polytronic, this translates to faster time-to-market for products, enhanced operational efficiency, and reduced infrastructure management costs.

### Continuous Integration and Continuous Delivery (CI/DC)

This implementation of CI/CD pipelines in cloud environments automates the testing, integration, and deployment cycles while maintaining high standards of quality and reliability for every release. By leveraging cloud native CI/CD tools such as AWS CodePipelining or Azure DevOps, Polytronic can

effectively adapt to customer needs, deploy updates seamlessly, and minimize downtime. This ensures an improved customer experience and enhances operational resilience.

## Financial Planning

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### **Infrastructure: Compute, Storage, and Networking**

Effective financial planning for cloud resources begins with allocating budgets strategically for compute, storage, and networking. Compute resources should be tailored to workload requirements, such as General-Purpose or Compute-Optimized instances to balance performance and cost. Scalable storage options like Amazon S3 buckets enable efficient data management, while budgeting for network costs ensures seamless data transfer across regions, which is essential for Polytronic's global operations.

### **Chargebacks: Resource Tagging**

Resource tagging allows businesses like Polytronic to track cloud resources costs accurately by department, project, or other organizational categories. By assigning tags to resources, financial accountability is promoted, enabling precise budgeting and forecasting. This practice supports a transparent chargeback system, ensuring each department is aware of and responsible for its resource consumption.

### **Instances: Reserved and Spot**

Polytronic can optimize cloud expenditures by using a combination of reserved and spot instances. Reserved instances are ideal for predictable and consistent workloads, providing cost savings through long-term commitments. For non-critical or flexible workloads, spot instances offer significantly reduced costs without compromising overall functionality. This dual approach ensures a balance between reliability and cost efficiency.

# Compliance and Security Framework – Compliance and Security

## Data Sovereignty

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For **Polytronic Training Systems LLC**, based in Abu Dhabi, UAE, data sovereignty impacts how data is stored and processed across different regions. Polytronic must ensure that its data management practices comply with the local laws and regulations of the UAE, as well as international regulations like the General Data Protection Regulation (GDPR) in Europe, China's Cybersecurity Law, and others depending on its global operations. The UAE has specific data protection laws that must be followed, particularly in sectors involving sensitive data. Polytronic must assess where its data is stored and processed to avoid potential legal penalties for cross-border data violations.

### How does data sovereignty affect your business?

Data sovereignty affects Polytronic by requiring compliance with local laws and regulations regarding data storage and processing. As Polytronic operates globally, it must ensure that data stored in the cloud or on local servers complies with specific regional laws, including the UAE Data Protection Law and international laws like GDPR in the EU. Challenges include navigating cross-border data movement restrictions, adhering to regional security standards, and implementing data protection measures like encryption and access control. Non-compliance could damage Polytronic's reputation and result in legal consequences. While adhering to these laws, may incur additional costs, (e.g., regional data centers, local compliance experts), compliance is essential for maintaining consumer trust, ensuring legal adherence, and safeguarding the business's global operations.

## Regulatory Concerns

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Regulatory Concerns refer to the issues, requirements, or challenges that an organization must address to comply with laws, standards, and regulations established by governing bodies or industry authorities. These concerns often involve security, privacy, environmental, financial, or operational standards, ensuring that an organization's activities align with legal and ethical expectations.

### How is Federal Risk and Authorization Management Program (FedRAMP) used by CSPs?

As a provider of cloud-based solutions to the federal government, Polytronic may partner with cloud Service Providers (CSPs) that are FedRAMP-certified to meet the federal government's stringent security standards. FedRAMP's standardized approach to security assessments, authorizations, a continuous monitoring ensures that cloud services meet federal government requirements. This makes it easier for Polytronic to deliver secure services to government clients.

### Who is responsible for adhering to FISMA requirements?

Polytronic's government contracts may require compliance with FISMA (Federal Information Security Modernization Act). If Polytronic stores or processes federal government data, it must ensure that both company and its contractors are FISMA-compliant by implementing the necessary security controls for federal systems.

### **How do FIPS protect sensitive information?**

Polytronic must adhere to FIPS (Federal Information Processing Standards) when handling sensitive government data, such as customer or financial information, FIPS 140-2 for instance, provides guidelines for the use of encryption to protect data both at rest and transit. Adhering to these standards ensures that sensitive data information managed by Polytronic is safeguarded in accordance with federal requirements.

## **Industry-Based Requirements**

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Industry-Based Requirements refers to the specific standards, guidelines, or best practices established by a particular industry to ensure consistent quality, safety, performance, and compliance among organizations within that sector. These requirements are often developed by industry associations, regulatory bodies, or standard-setting organizations and are designed to address the unique needs and challenges of the industry.

### **How is FERPA similar to HIPAA?**

FERPA (Family Educational Rights and Privacy Act) and HIPAA (Health Insurance Portability and Accountability Act) both emphasize the importance of protecting personal information. FERPA safeguards the privacy of student education records, while HIPAA protects health-related information. Polytronic may handle educational data through its educational technology services or healthcare data via health-related solutions, necessitating compliance with both regulations.

### **How does FINRA ensure financial institutions retain important transactions and communication?**

If Polytronic provides cloud services to financial institutions, it must comply with FINRA (Financial Industry Regulatory Authority) requirements. FINRA mandates that financial firms retain transactions records and communications for regulatory purposes. Polytronic must ensure its solutions are capable of securely storing this data and making it available to regulators when needed.

### **How does GLBA protect the customers or your business?**

Polytronic must comply with the GLBA (Gramm-Leach-Bliley Act) if it stores or processes financial customer data. This act mandates that financial institutions protect customer's non-public information (NPI). Polytronic should implement robust security practices, such as encryption, secure access controls, and continuous monitoring, to safeguard customers' data effectively.

### **Why is SOX important to publicly traded companies?**

The Sarbanes-Oxley Act (SOX) is important to publicly traded companies including Polytronic, as it ensures the accuracy and reliability of financial reporting, which is vital for maintaining investor trust and confidence. By establishing strict internal controls, requiring independent audits, and holding executives personally accountable for the accuracy of financial statements, SOX plays a crucial role in preventing corporate fraud and financial mismanagement.

### **Why is security main focus with MPAA?**

As a company that may provide digital media services, Polytronic must adhere to MPAA (Motion Picture Association of America) security standards to protect copyrighted films and media content. These standards help prevent piracy and unauthorized distribution by requiring encryption, watermarking, and secure access to methods to ensure that media assets are not compromised.

## **International Standards**

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International Standards refer to globally recognized rules, guidelines, or specifications established to ensure quality, safety, efficiency, and interoperability across different industries, sectors, or regions. These standards are typically developed and maintained by international organizations such as International Organization for Standardization (ISO), the International Electrotechnical Commission (IEC), or the International Telecommunication Union (ITU). They aim to facilitate global trade, enhance product compatibility, and ensure uniformity in practices across borders.

### **How does ISO help CSPs and businesses?**

Polytronic can benefit from adhering to ISO standards, such as ISO 27001, which provides a framework for establishing, implementing, and maintaining an Information Security Management System (ISMS). By aligning with these international standards, Polytronic ensures that it meets global expectations for data protection, mitigate risks, and enhances trust with international customers.

## **Certifications**

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Certification refers to the formal process by which an individual, organization, product, or service is evaluated and verified against the specific standards, requirements, or criteria established by a recognized authority or certifying body. The goal of certification is to demonstrate compliance, competence, or quality in a particular area, ensuring trust, credibility, and accountability.

### **Describe the principles within the CIA Triad.**

The CIA Triad is a foundational concept in information security, representing three core principles that guide the protection of data and systems. In Polytronic's cloud operations, the CIA Triad – Confidentiality, Integrity, and Availability guides all security practices.

**Confidentiality:** ensures that only authorized users can access sensitive data.

**Integrity:** ensures that the data remains unaltered and accurate throughout its lifecycle.

**Availability:** ensures that data and systems are accessible when authorized users need them.

### **Share an example of what a threat is and how to identify one in your business.**

A common threat for Polytronic could be a phishing attack, where attackers attempt to steal sensitive data, such as login credentials. Polytronic can identify this threat by monitoring email traffic for suspicious messages that attempt to trick users into revealing personal information. Security awareness training for employees can also help detect and prevent such attacks.

### **What is the difference between a threat and vulnerability in your business?**

In Polytronic, a **threat** refers to any potential event that could harm the business or compromise security (e.g., cyberattacks, data breaches). A **vulnerability** is a weakness or gap in Polytronic's security posture that could be exploited by the threat. For example, weak passwords or outdated software are vulnerabilities, and an attacker exploiting those vulnerabilities is the threat.

### **What are the primary tools for a security assessment that you would use in your business?**

Polytronic can use a combination of vulnerability scanning tools like Nessus, or Qualys, penetration testing tools such as Metasploit, and security auditing software like Wireshark and Nmap. To assess the security posture of its cloud systems and infrastructure.

### **What are the three categories for security and their meaning in relation to your business?**

Polytronic should focus on:

**Data Rest:** Data stored in Polytronic's cloud infrastructure must be encrypted and protected from unauthorized access.

**Data in Transit:** Data moving across network, which should be secured using encryption protocols like SSL/TLS.

**Data in Use:** Data being processed, which should be secured through real-time monitoring and access control.

### **Describe the difference between access and authorization.**

**Access** refers to a user's ability to interact with a system or resources, while **authorization** is the specific permission that determines what a user can do within the system. For example, a user might have access to database but may not be authorized to modify certain records, depending on their role.

## **Security Concerns, Measures, and Concepts**

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In any cloud proposal, security is vital. **Polytronic Training Systems LLC** plans to adopt robust measures to address potential security threats. Key security measures include:

**Encryption:** All sensitive data should be encrypted at rest (e.g., AES encryption) and in transit (e.g., HTTPS).

**Access Controls:** Role-based access control (RBAC) to restrict access to sensitive systems and data.

**Multi -Factor Authentication (MFA):** To prevent unauthorized access to Polytronic's systems, especially in cloud environments.

**Regular Security Audits:** Conducting frequent audits to detect vulnerabilities and ensure compliance with industry standards and regulations (ISO 27001 or SOC 2).



# Risk Management and Governance

## Strategy – Governance and Risk

### Risk Assessment

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Risk assessment evaluates potential threats and vulnerabilities that could harm the business.

#### Basic Assets of Value:

**Tangible:** Physical items such as servers, laptops, and office space essential for operations.

**Intangible:** Non-physical items like company reputation, customer trust, and proprietary software.

#### Quantitative Risks:

These are non-measurable risks assessed based on likelihood and impact, such as:

Data breaches (e.g., hacking incidents)

Insider threats (e.g., employee misuse of access)

Service outages (e.g., cloud provider downtime)

#### Categorize Assets:

**Tangible:** Physical objects like hardware or equipment

**Intangible:** Abstract items like intellectual property, brand value, and customer loyalty.

#### Risk Classification (Scale 1-5)

Each is rated based on severity (5 = critical, 1 = low):

Data breach: 5 - Server financial and reputational damage.

Insider threats: 4 – Significant damage but manageable with controls.

Service outage: Moderate disruption but temporary.

#### Risk Owner and Asset Owner:

**Risk Owner:** The Risk owner is responsible for managing and mitigating the identified risk. For each risk, the owner would be:

- Data Breach: Chief Security Officer (CSO)
- Insider Threats: IT Security Manager
- Service Outage: Cloud Service Manager

**Asset Owner:** The asset owner is responsible for ensuring the security and integrity of the asset. For each asset, the owner would be:

- Data (Customer Data, Training Content): Data Protection Officer (DPO)
- Intellectual Property: Chief Technology Officer: (CTO)
- Brand Reputation: Marketing and PR manager

## Risk Response

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This is how a business prepares for, and addresses identified risks.

### **What It Means:**

Risk response involves deciding on strategies to reduce, transfer, accept, or avoid risks.

### **Mitigation Strategies:**

Deploy firewalls to prevent unauthorized access.

Train employees to recognize phishing attempts.

Ensure backup systems for business continuity during outages.

### **Risk Acceptance:**

Some risks, like minor service interruptions, may be accepted if mitigation costs outweigh benefits.

### **Risk Avoidance:**

Choosing not to engage in activities with high risks, like using insecure cloud services, is the safest response.

### **Risk Transference (via CSP Service):**

When a Cloud Service Provider (CSP) is utilized, certain risks such as physical data center security and uptime reliability, to the provider.

## Documentation

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Documentation is the backbone of IT operations, as it helps maintain consistency, track changes, and ensure compliance.

### **Why it's important:**

Keeps systems standardized.

Facilitates troubleshooting and failure planning.

### **Examples of Findings:**

Outdated Password Policy: Leads to weak system defenses.

Unpatched Software: Vulnerable to known exploits.

## Vendor Lock-In

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Vendor Lock-In refers to a situation where a business becomes heavily dependent on a specific service provider (CSP) due to the use of proprietary technologies, or the high costs associated with switching to another provider.

### **What Is and Why It is a Risk:**

Vendor lock-in limits flexibility by making it costly or technically challenging to migrate data or services.

For instance, if Polytronic uses a proprietary database from a CSP, moving to another provider might require significant redevelopment.

### **Data Portability vs. Vendor Lock-In:**

**Data Portability:** The ease of transferring data between platforms.

**Vendor Lock-In:** The inability of migrate due to technical or financial barriers.

## **Policies and Procedures**

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Policies and Procedures refer to a structured framework of rules, guidelines, and processes designed to ensure the consistent and effective management of data, systems, and operations within an organization.

### **Data Privacy:**

Protecting customer and business data from unauthorized access or misuse is essential for compliance and customer trust (e.g., GDPR, HIPAA)

### **Examples of Policies:**

Data Retention Policy: Defines how long data is kept before disposal.

Incident Response Policy: Details steps for detecting and addressing security incidents.

### **Change and Resource Management:**

Ensures that changes to IT infrastructure (e.g., upgrades or new deployments) are planned and resources are allocated efficiently, reducing risks of disruption.

### **Incident Response Steps:**

Detecting the incident.

Contain the damage.

Eradicate the threat.

Recover and document findings.

### **Three Basic ACL Rights:**

**Read:** Permission to view files or systems.

**Write:** Permission to modify files or systems.

**Execute:** Permission to run programs or scripts.